Katherine Titterton

Scientist-engineer and technical storyteller in neuroscience, imaging and AI

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kat-titter

Analytical researcher in high-content imaging. Develops and scales assays until the signal is clean, then extracts what the images hold: from *in vitro* neurodegeneration models to AI-ready datasets. Learned to code at night before AI could; now builds with people and AI agents, for people.

EXPERIENCE

Axiom Bio Founding Member of the Technical Staff — San Francisco, CA — 2023–2026

- Oversaw generation and QC of 10M+ Cell Painting images of 100,000+ compounds (nearly every commercial bioactive), the training data for models that predict bioactivity profiles from structure alone, before synthesis.
- Invented two low-noise algorithms for morphology change, ~10× more sensitive than conventional cytotoxicity assays, which made them top features of a 0.92 AUC liver-injury (DILI) model; they also caught batch drift.
- Led image labeling and U-Net segmentation, in production for 1+ year.

  Nat Commun 2026 [4] Cell Syst 2026 [3] bioRxiv 2026 [P1]

Insitro Associate Scientist III, Arrayed Screening — South San Francisco, CA — 2022–2023

- Adapted and ran high-content phenotypic screens for tuberous sclerosis and ALS on automated platforms.
- Automated QC and analysis of hundreds of 384-well plates with Python/R scripts.
- Built a digital phase-contrast imaging modality and segmentation for label-free tracking of neuron growth.

MIT Technical Associate II, Synthetic Neurobiology Group — Cambridge, MA — 2021–2022

- Optimized multiplexed expansion revealing (mExR) for 7+ immunostaining rounds: 30+ proteins at nanoscale.

  Nat Commun 2024 [6]

AbbVie Associate Scientist II, Foundational Neuroscience Center — Cambridge, MA — 2018–2021

- Cultured primary neurons weekly for 3+ years, supporting 10+ tau team scientists; scaled the culture from manual to automated by onboarding Integra liquid handlers.
- Developed high-content tau aggregation assays in primary and iPSC neurons to rank immunotherapies.
- Tested next-gen microfluidic devices, keeping primary murine neurons alive 3+ months to model tau spread.

  Mol Neurodegener Adv 2026 [1] iScience 2026 [2] J Biol Chem 2025 [5] iScience 2022 [7]

Massachusetts General Hospital Research Technician I — Boston, MA — 2017–2018

- Recorded voltage-clamp currents from GABA-A receptors in *Xenopus* oocytes to study how anesthetics act.

  Eur J Med Chem 2019 [8]

PROJECTS

Datum Line AI × bio startup: one shared map of every lab's controls, to tell if cells look normal — 2026
Every line is a loop Does biology have a periodic table? Interactive essay — 2026
Overdose Watch Public tracker a weekly Claude Code agent keeps current — 2026
Vanitas / Veritas AI slop as vanitas: 4K open-source diffusion loop, interactive piece — 2026
claude-code-learns Graph-engineered agent harness: corrections become skills, auto-checked — 2026
Vivarium 2.0 Automated poison-dart-frog habitat with misting and live video — 2024
OpenTronsformers LLM copilot for the OT-2 lab robot; 2nd-place team, BioHack NYC — 2023

SKILLS

Code & agents: Python, R, Git, Claude Code (skills, hooks, scheduled agents), agent graph engineering, evals
ML & data: PyTorch, U-Net and SAM segmentation, image embeddings, feature engineering, signal extraction
High-content screening: primary and iPSC neurons, immunocytochemistry (ICC), immunohistochemistry (IHC)
Lab automation: Integra, BioTek, HighRes and Opentrons liquid handlers and schedulers, assay miniaturization
Story & design: data visualization, figure, deck and web design, generative art (Midjourney, diffusion)

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PUBLICATIONS

[1] Yanamandra K, …, **Titterton K**, …, Langlois X. Conformation-specific anti-pathological tau antibody raised against AD brain-derived seed-competent tau reduces tau pathology in transgenic mice. *Mol Neurodegener Adv*. 2026;2(1):43. doi:10.1186/s44477-026-00049-6 ABBVIE

[2] Sebastian RM, …, **Titterton K**, …, Wu JW. SynCAR platform for capturing neuronal synaptic proteopathic seeds. *iScience*. 2026;29(7):116362. doi:10.1016/j.isci.2026.116362 ABBVIE

[3] Ewald JD, **Titterton KL**, …, Singh S. Cell Painting for cytotoxicity and mode-of-action analysis in primary human hepatocytes. *Cell Syst*. 2026;17(5):101566. doi:10.1016/j.cels.2026.101566 AXIOM BIO

[4] Seal S, …, **Titterton K**, …, Carpenter AE. Counting cells can accurately predict small-molecule bioactivity benchmarks. *Nat Commun*. 2026;17(1). doi:10.1038/s41467-026-68725-5 AXIOM BIO
featured on axi.om

[5] Wu JW, **Titterton K**, …, Langlois X. Development of a robust, physiologically relevant neuronal tau aggregation assay for tau-related drug discovery. *J Biol Chem*. 2025;301(12):110853. doi:10.1016/j.jbc.2025.110853 ABBVIE

[6] Kang J, …, **Titterton K**, …, Boyden ES. Multiplexed expansion revealing for imaging multiprotein nanostructures in healthy and diseased brain. *Nat Commun*. 2024;15(1):9722. doi:10.1038/s41467-024-53729-w MIT

[7] Manos JD, …, **Titterton K**, …, Langlois X. Uncovering specificity of endogenous TAU aggregation in a human iPSC-neuron TAU seeding model. *iScience*. 2022;25(1):103658. doi:10.1016/j.isci.2021.103658 ABBVIE

[8] Wu B, …, **Titterton K**, …, Bruzik KS. Inhibitable photolabeling by neurosteroid diazirine analog in the β3-subunit of human heteropentameric type A GABA receptors. *Eur J Med Chem*. 2019;162:810-824. doi:10.1016/j.ejmech.2018.11.019 MGH

PREPRINT

[P1] Ollitrault G, …, **Titterton K**, Ewald J. Cell Painting and machine learning for mechanistic profiling of hepatotoxic botanicals and their constituents. *bioRxiv*. 2026. doi:10.64898/2026.10.05.756246 AXIOM BIO

SELECTED CONFERENCE PRESENTATIONS & POSTERS

[1] **Titterton K**. Using AI to Combine Exposure, Metabolism, and Liver Function to Model DILI in Modern Molecules. Talk, American College of Toxicology (ACT) Annual Meeting, Phoenix, AZ, 2025. replay

[2] **Titterton K**, Boiko D, Cabrera ÁA, Bäuerle A, Tyagi A, Saneinejad S, Beatson A, White B. Accurate and interpretable *in silico* clinical risk assessment for drug-induced liver injury (DILI) from molecular structure. *Society of Toxicology Annual Meeting*, 2025. poster

[3] **Titterton K**, et al. Development of a tau propagation neuronal microfluidic cell culture model for assessing inhibitors of tau propagation. *Society for Neuroscience Global Connectome*, 2021. poster

[4] **Titterton K**, et al. High content analysis of tau aggregation in primary neurons: screening anti-tau antibodies and modeling tauopathy progression *in vitro*. *SBI2 High Content Meeting*, Harvard Medical School, 2019. poster

EDUCATION

Colgate University B.A. Cellular Neuroscience, minor Mathematical Systems Biology 2013–2017

Harvard Extension School Coursework 2017–2020

Undergraduate and graduate: Python, computer vision, social medicine, biochemistry.